Quantifying the Resource Cost of Shor's factoring algorithm

Guangqi Gao, Jerry Xia, Avery Adam Books, Yiqun Zhao
Department of Physics, University of California, Berkeley

Introduction

Shor's algorithm is a quantum algorithm that efficiently factors large integers—an essential task underlying the security of widely used cryptographic systems such as RSA. Classically, factoring requires **super-polynomial time** in the worst case, but Shor's algorithm reduces this to **polynomial time** by using a quantum subroutine for order finding.

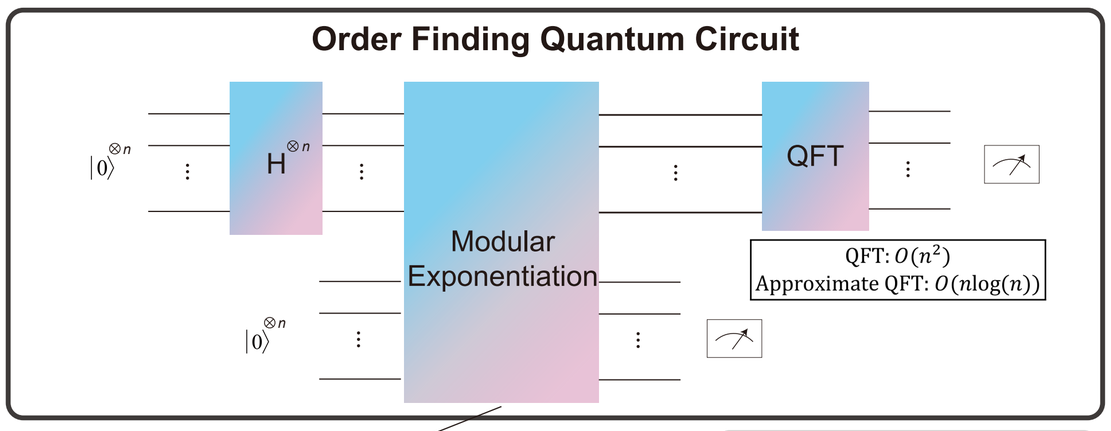
The algorithm combines simple classical preprocessing with a quantum circuit that determines the order r of a modulo N.

$$r = \min\{k > 0 : a^k \equiv 1 \pmod{N}\}.$$

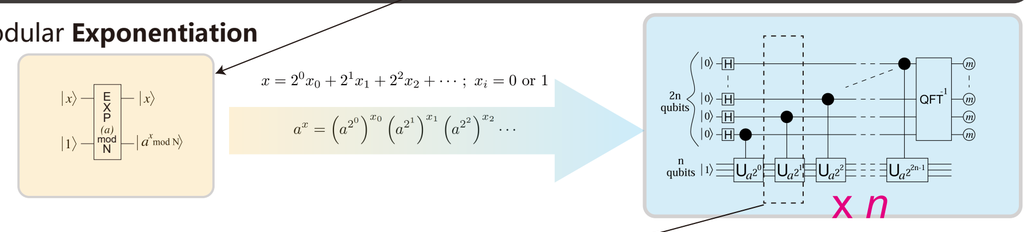
Once r is found, classical postprocessing retrieves nontrivial factors of N. The quantum speedup arises entirely from this order-finding step, which relies on modular exponentiation and the Quantum Fourier Transform.

In this work, we first review the theoretical foundations of universality, and analyze the asymptotic complexity of each arithmetic component of order finding circuit. We also estimate how these resource requirements scale under surface-code quantum error correction.

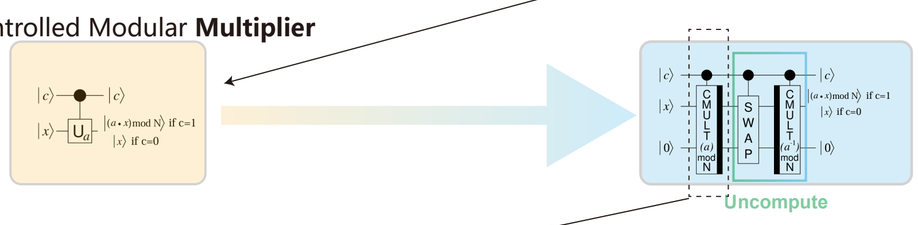
Complexity of Order Finding Quantum Circuit



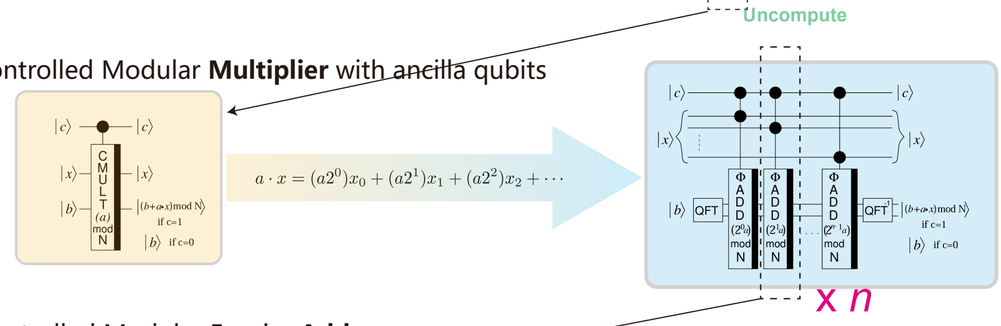
5. Modular **Exponentiation**



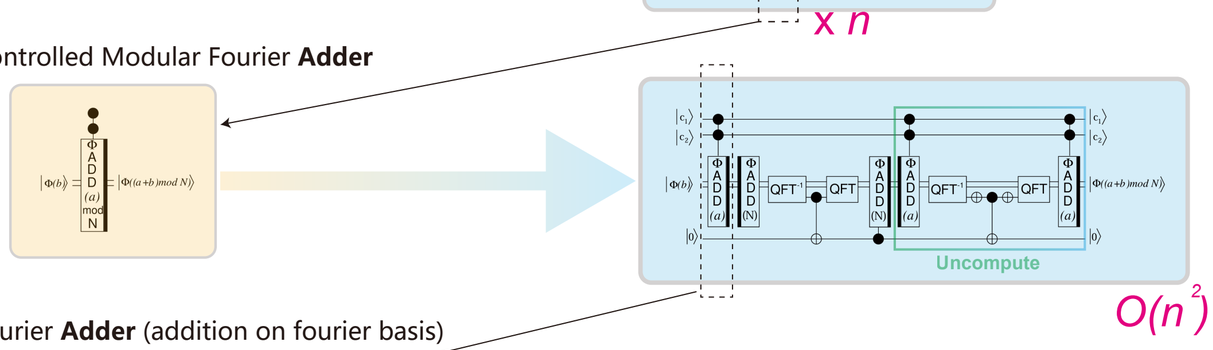
4. Controlled Modular **Multiplier**



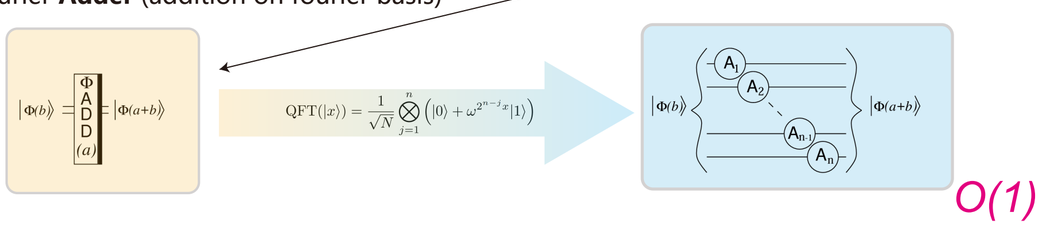
3. Controlled Modular **Multiplier** with ancilla qubits



2. Controlled Modular Fourier **Adder**



1. Fourier **Adder** (addition on fourier basis)



High-level Circuit

Decomposited Circuit

The quantum part of Shor's algorithm implements order finding by decomposing the computation into two major components: the Quantum Fourier Transform (QFT) and modular exponentiation. The QFT, which recovers the hidden period from the control register, requires $O(n^2)$ gates, or $O(n\log(n))$ when using the approximate version. Modular exponentiation, which computes $a^x \bmod N$ reversibly, is built from Fourier adders, modular adders, and controlled modular multipliers, with respective complexities $O(n)$, $O(n\log(n))$, $O(n^2\log(n))$. Repeated use of these arithmetic blocks leads to an overall gate complexity of $O(n^3\log(n))$, making modular exponentiation the dominant cost in the quantum order-finding circuit.

Universal gate sets

Arbitrary 1-Qubit Gates + One Entangling 2-Qubit Gate = Universality

- $\{H, T, \mathrm{CNOT}\}$ — the standard Clifford+T set used in fault-tolerant quantum computing
- $\{R_Z(\theta), R_X(\theta), \mathrm{CNOT}\}$ — common in analog quantum control
- $\{U_3, \mathrm{CX}\}$ — the default universal basis for IBM Quantum systems
- $\{R_Z, SX, X, \mathrm{CX}\}$ — minimal and hardware-native set also known to be universal

Gate-Set Independence of BQP

The Solovay–Kitaev theorem guarantees that any such approximation can be achieved with circuit depth scaling as $\mathcal{O}(\log^c(1/\epsilon))$, for $c < 4$.

Gate Set Selection for This Project

- $R_Z(\theta)$ is implemented virtually, incurring no gate time or decoherence.
- SX and X are single-qubit rotations about the X axis.
- CX is the standard two-qubit entangling operation.

Toward Fault-Tolerant Universality: Clifford+T Compatibility

$$R_Z(\theta) \xrightarrow{\text{approx.}} \text{Clifford} + \text{T}$$

We prioritize the gate set $\{R_Z, SX, X, \mathrm{CX}\}$ for its experimental feasibility, low overhead, and decomposition efficiency. At the same time, we ensure compatibility with Clifford+T constructions, recognizing their role in error-corrected architectures. This layered approach allows us to bridge practical implementation with long-term scalability in quantum computing.

Extra Cost of Quantum Error Correction

The industry standard is 2D surface codes implemented on super-conducting circuits demonstrated in Google and IBM's current quantm chips.

In order for CX gates to be implemented in a surface code, lattice surgery must be used to circumvent nearest neighbor connectivity restrictions.

The QEC Cycle is as follows:

1. Ancillary qubits set to ground state of respective basis.
2. 4 CX gates are performed between measurement qubit and four nearest neighbor qubits.
3. Measurement for X errors is done via 4 Z-stabilizers depending on the syndrome produced by the CX gates.
4. Syndromes are tracked via classical software and corrections are made at the end.

QEC also increases the time and spatial complexity of a system. Our paper estimated the number of qubits required (spatial copmlexity) of implementing a surface code on a Shor algorithm with 50% space for T factories to be of order **N*log(N)^2** where N is the number of logical qubits.

References

[1] A. Barenco et al., Elementary Gates for Quantum Computation, Phys. Rev. A 52, 3457–3467 (1995)
[2] S. Beauregard, Circuit for Shor's Algorithm Using 2n+3 Qubits, arXiv:quant-ph/0205095 (2003).
[3] C. Gidney and M. Ekerå, How to Factor 2048-bit RSA Integers in 8 Hours Using 20 Million Noisy Qubits, Quantum 5, 433 (2021).

Team contributions

G. Gao & J. Xia reconstructed and analyzed the quantum order finding circuit. Y. Zhao summarized universal gate set foundations, and A. Books analyzed QEC overhead using surface-code methods.